Trait-based Ecology IV

Today's Agenda:

- Quiz
- Traits and global change

Why focus on traits?

- Generality:
  - Rather than focusing on particular species in particular systems, we can say things about all species in all systems (or at least a larger subset of species/systems).
- Quantitative:
  - We move from species (categorical) to continuous measurements
- Links to mechanism
  - We can focus on traits that are linked to particular mechanisms
  - E.g., if we want to work on predation, mouth size might be useful.
- More predictive
  - Allow us to say something about potential future scenarios

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Traits and global change

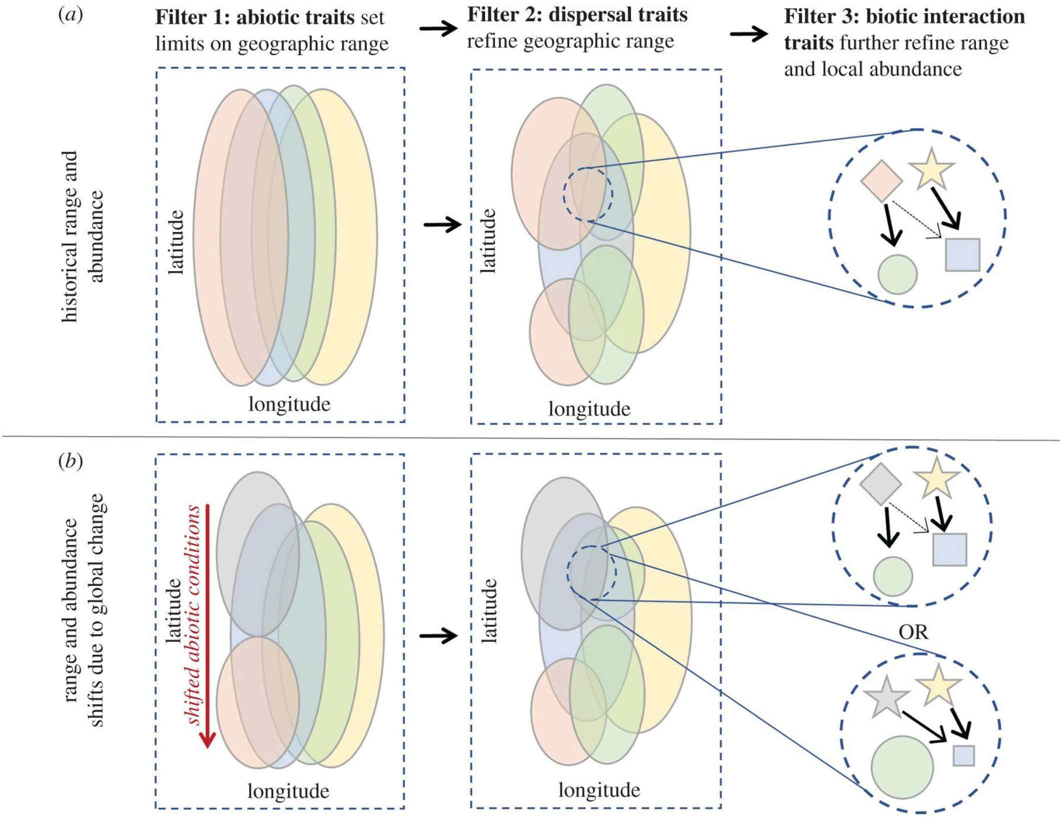
- Overview
- What traits?
- How will species respond to climate change?
- Fire regime shifts?
- Which species are humans impacting?

Traits and global change

PROCEEDINGS B
royalsocietypublishing.org/journal/rspb
Review

Trait-based approaches to global change ecology: moving from description to prediction

Stephanie J. Green[1], Cole B. Brookson[1], Natasha A. Hardy[1,2] and Larry B. Crowder[2]



ellipse size/position = species geographic range; colour = species identity; shape = trait set; shape size = relative abundance; arrow thickness = strength of consumptive interaction within local food web.

Traits and global change

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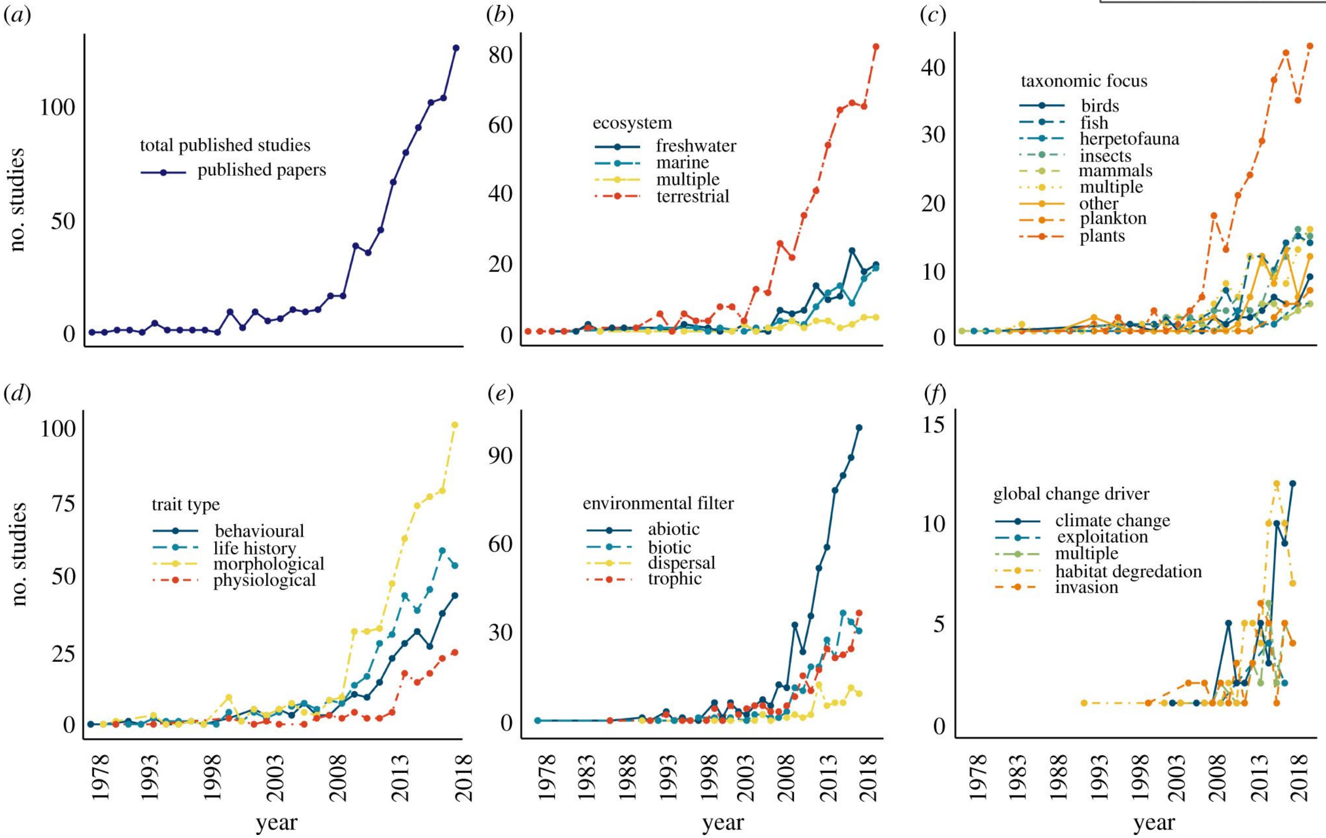

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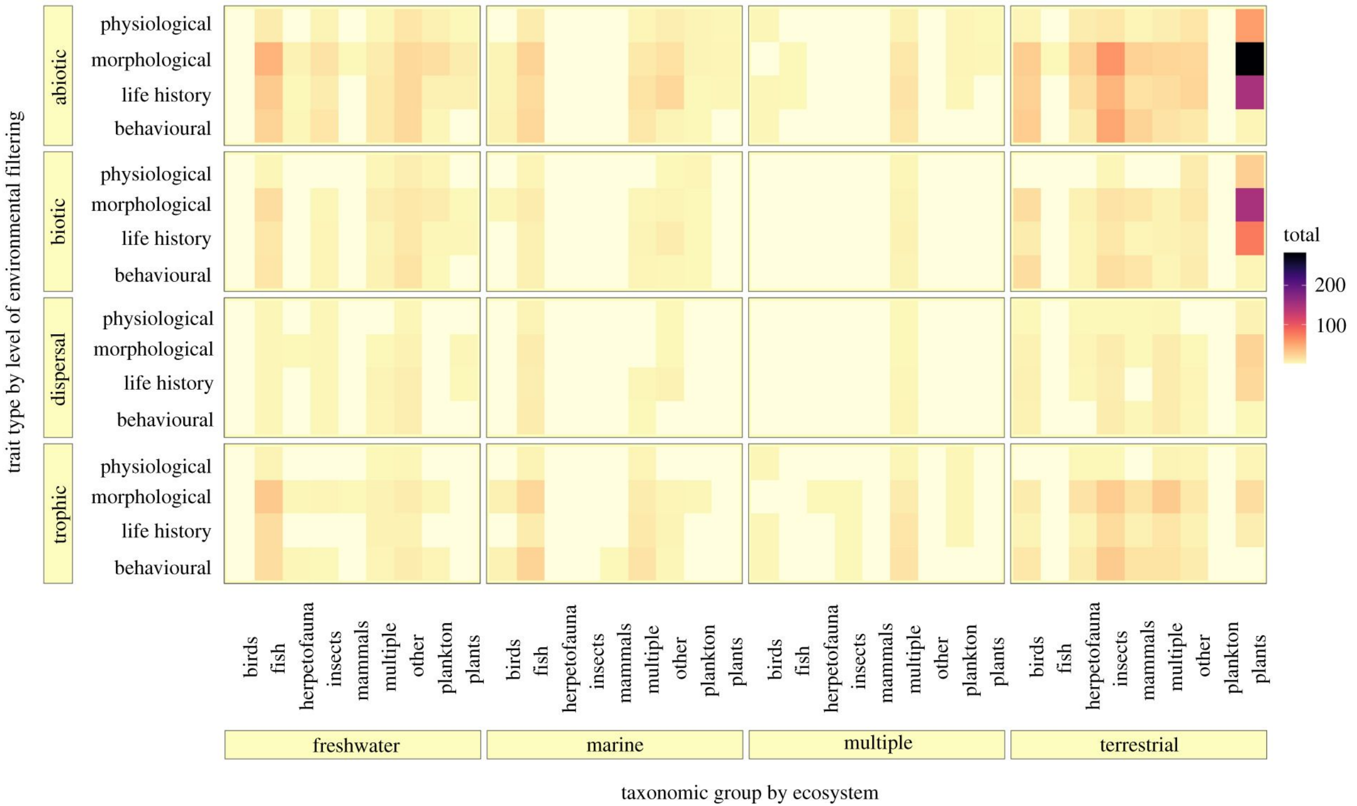

What traits should we measure?

Literature review of 10 years worth of papers

Frontiers of Biogeography 2021, 13.04, e53774

REVIEW

Frontiers of Biogeography
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Globally important plant functional traits for coping with climate change

Nicola Kühn[1,2], Carolina Tovar[1], Julia Carretero[1], Vigdis Vandvik[3,4], Brian J. Enquist[5,6] and Kathy J. Willis[1,7]

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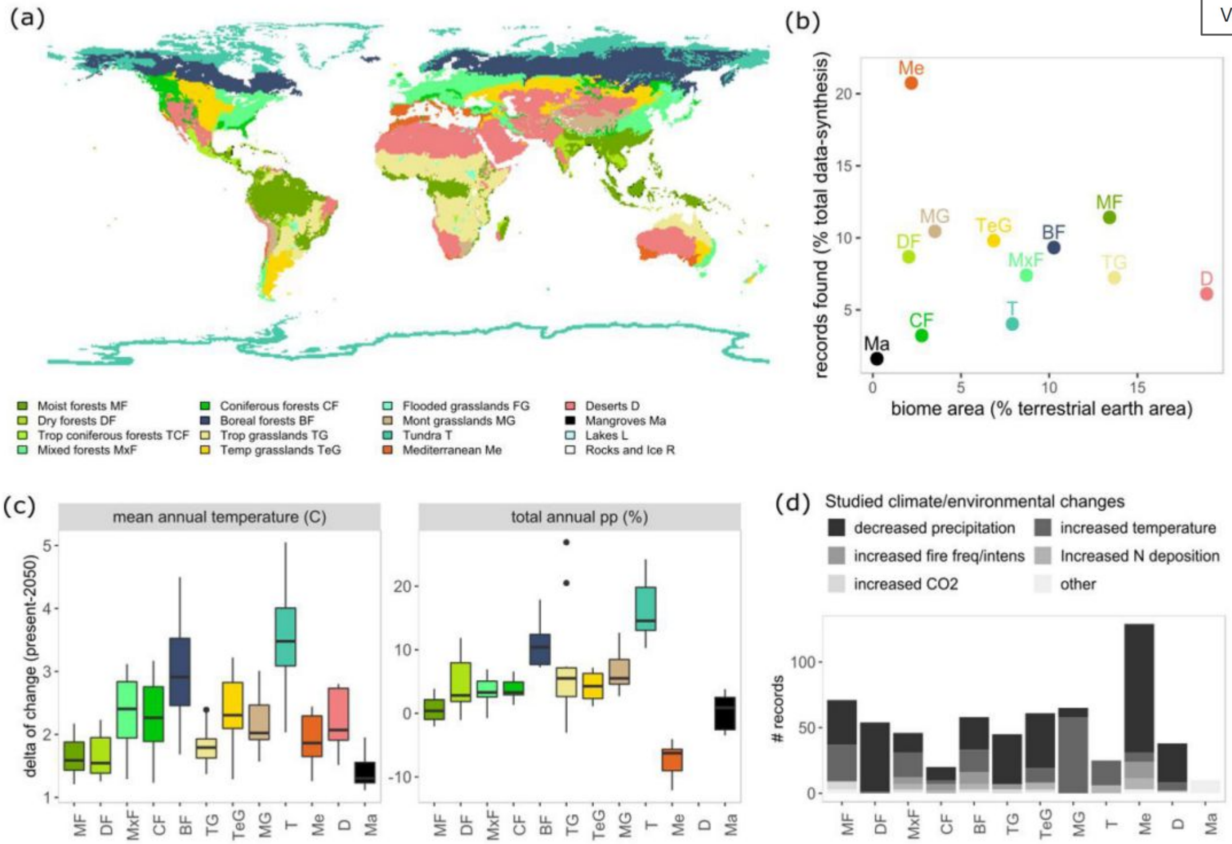
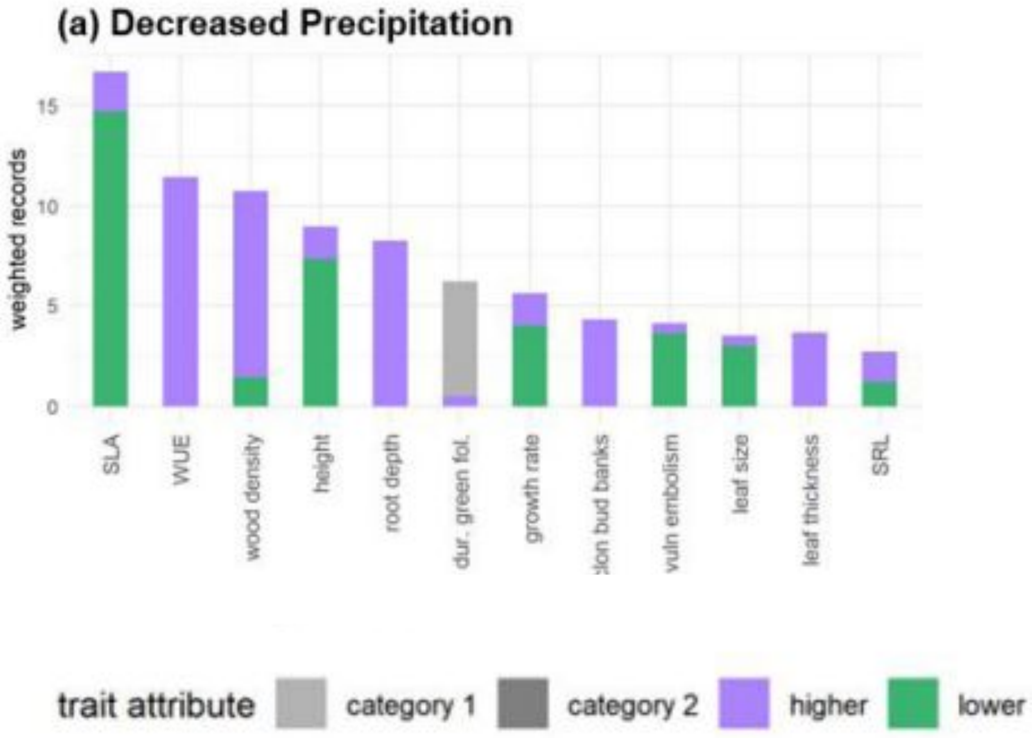


Biases in what we're studying

What traits should we measure?



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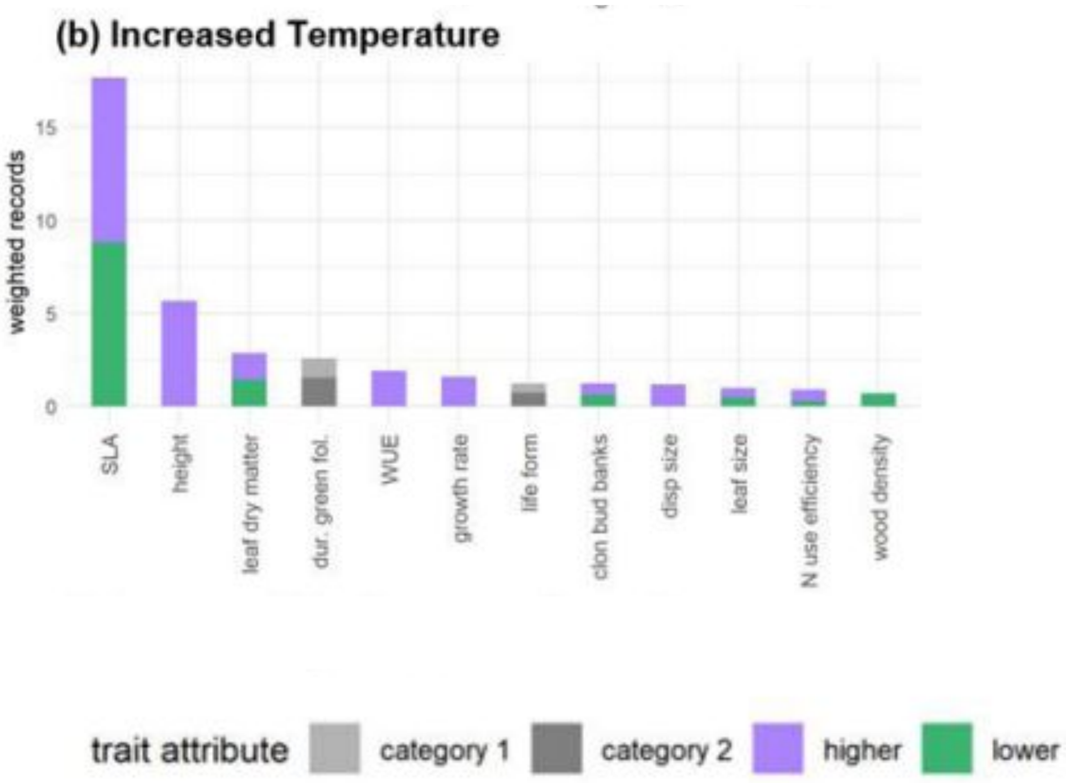

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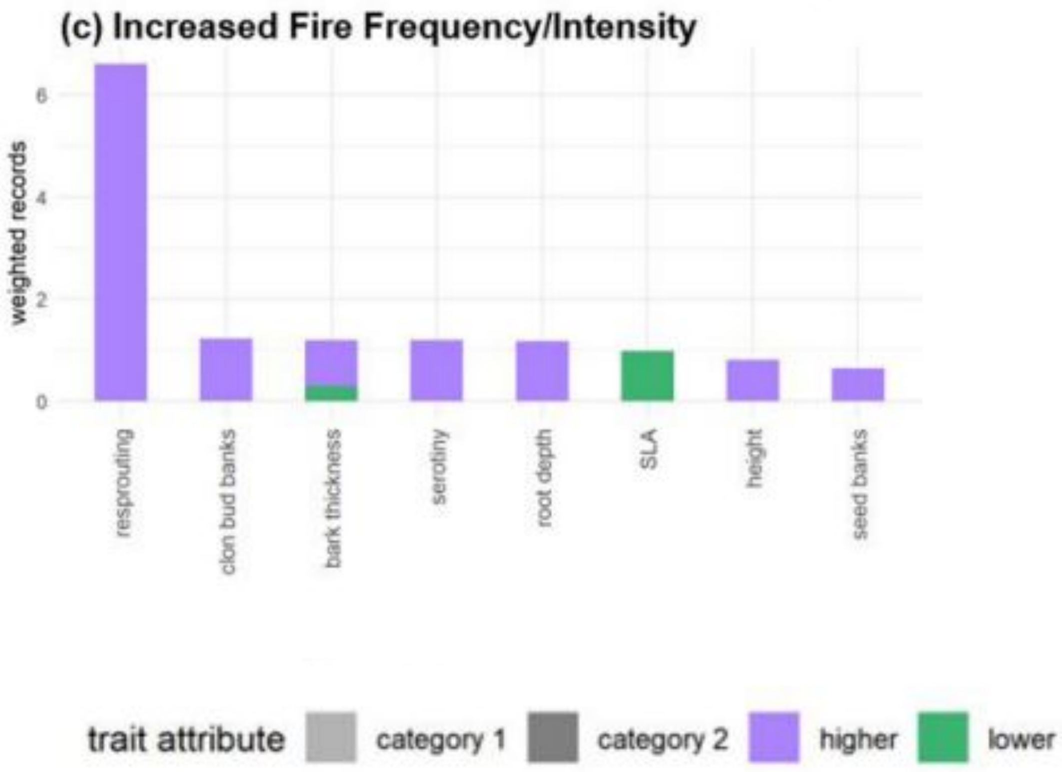

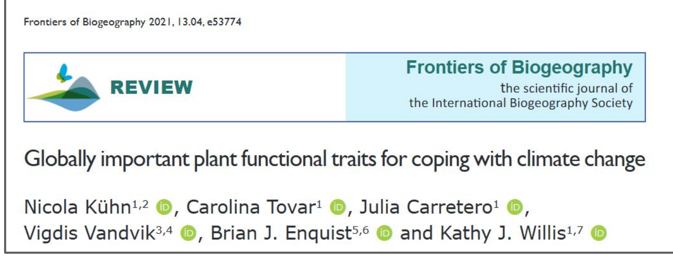
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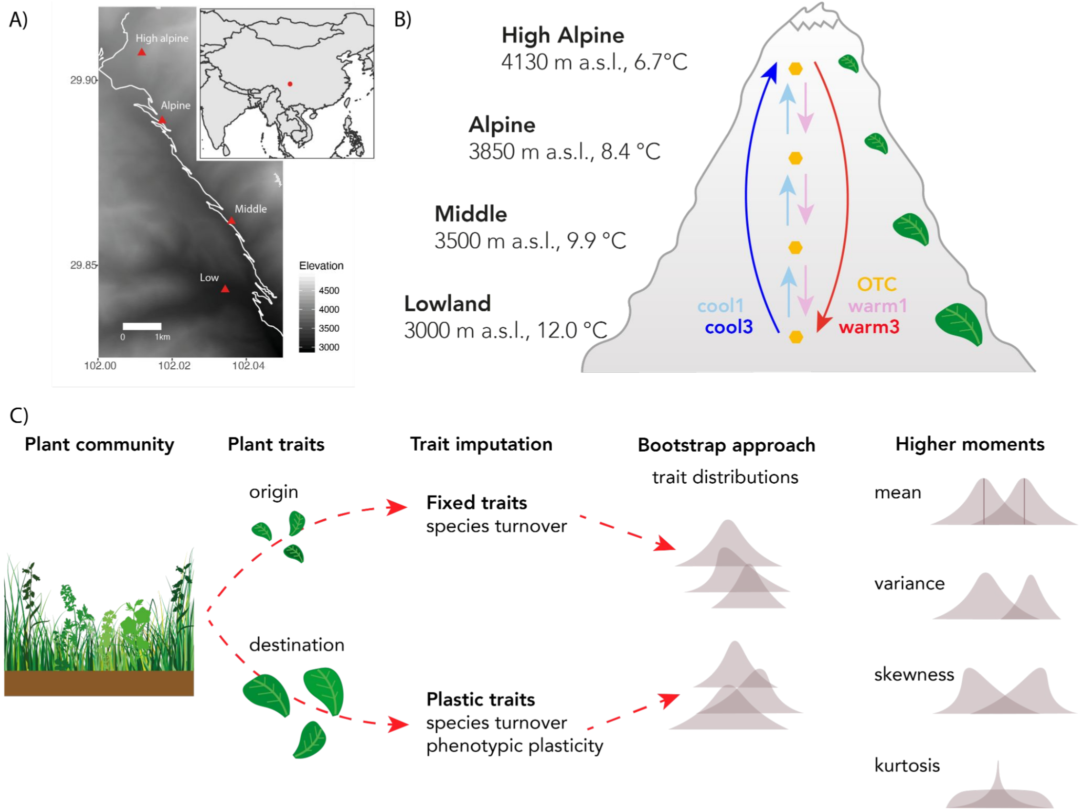
Predicting traits



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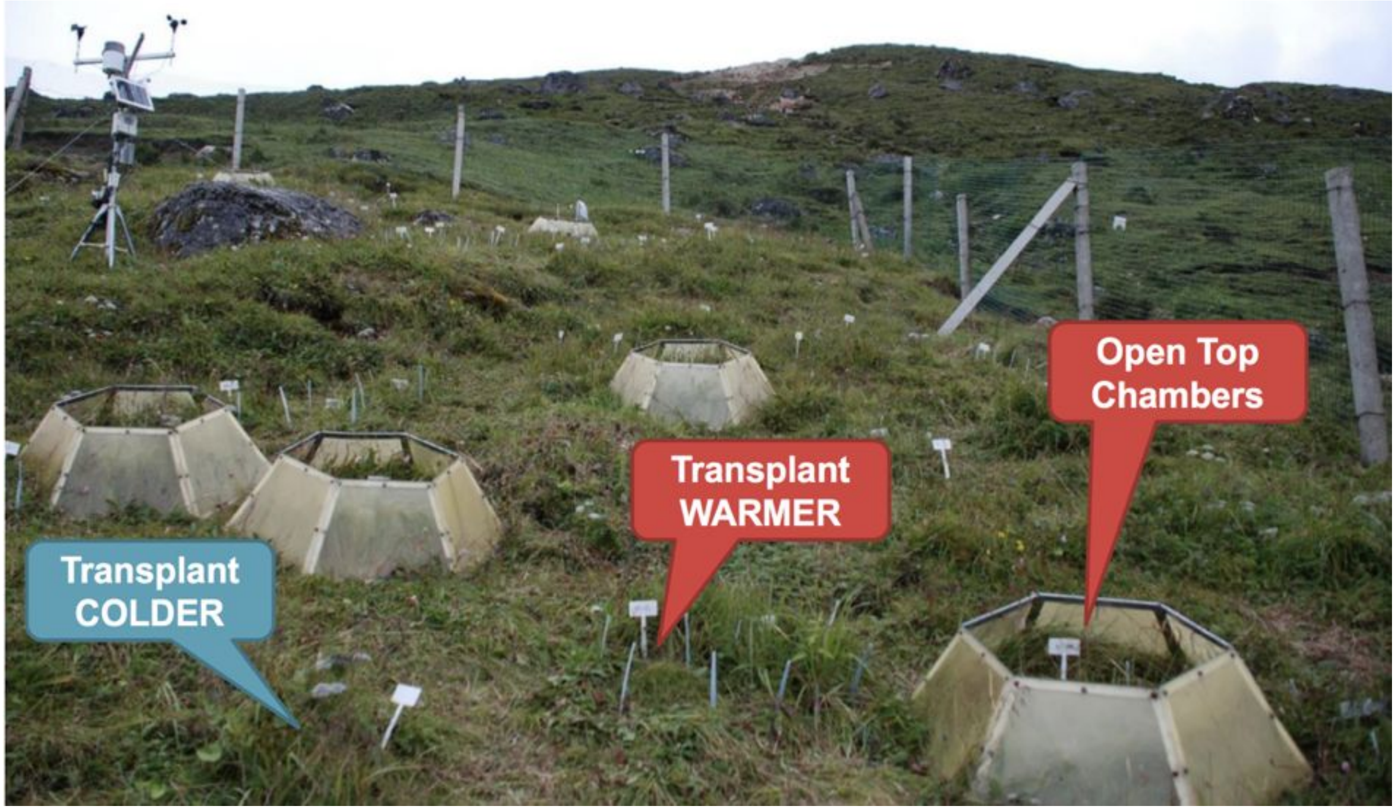


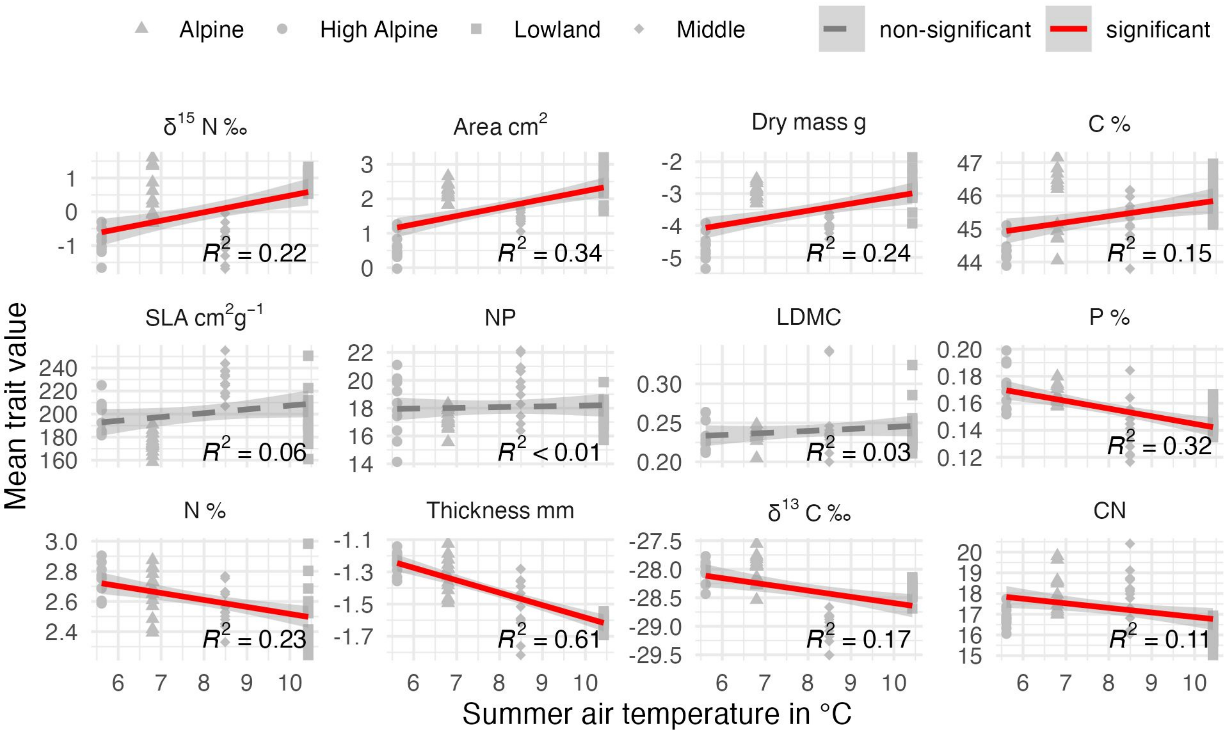
Predicting traits

Compared three approaches:

- Gradients
- Transplants
- OTCs

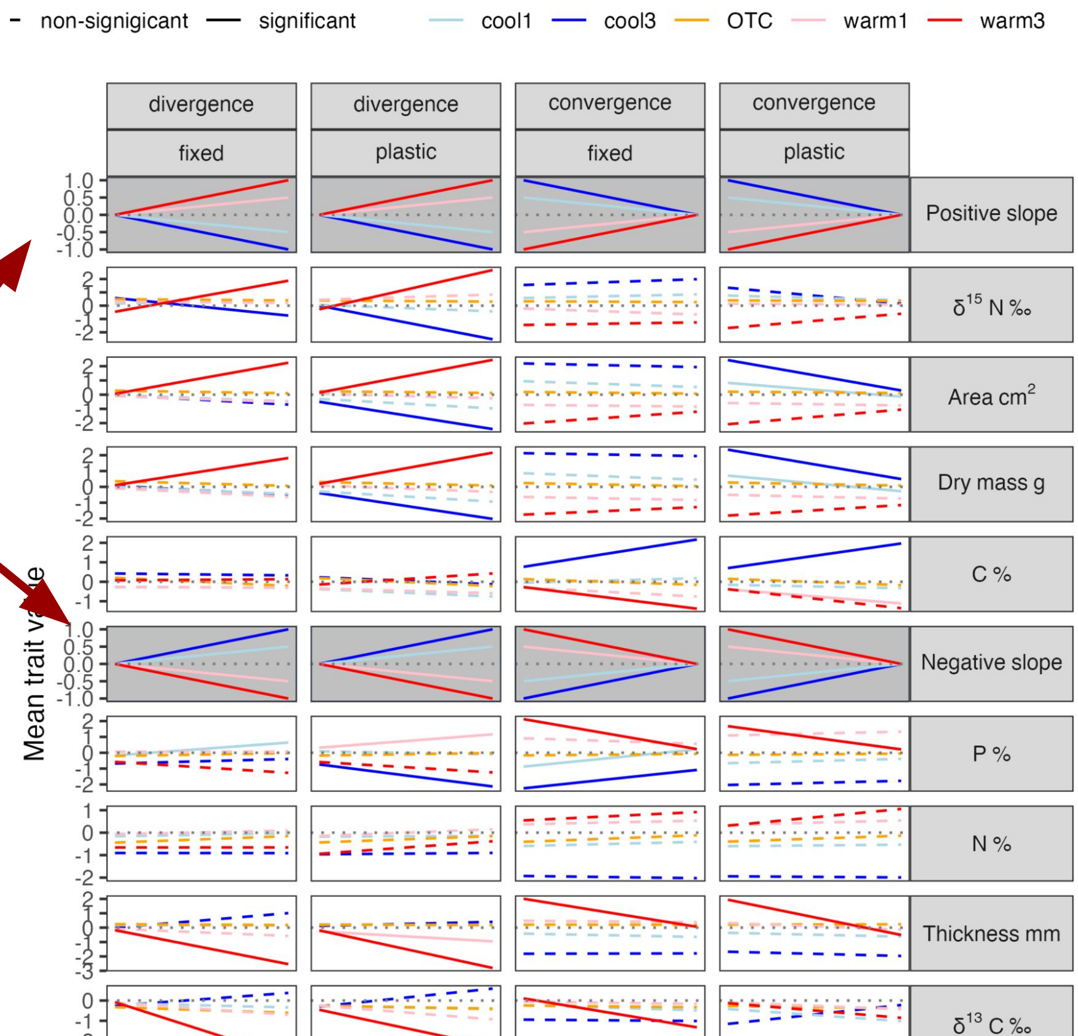
Predicting traits

Gradient patterns:

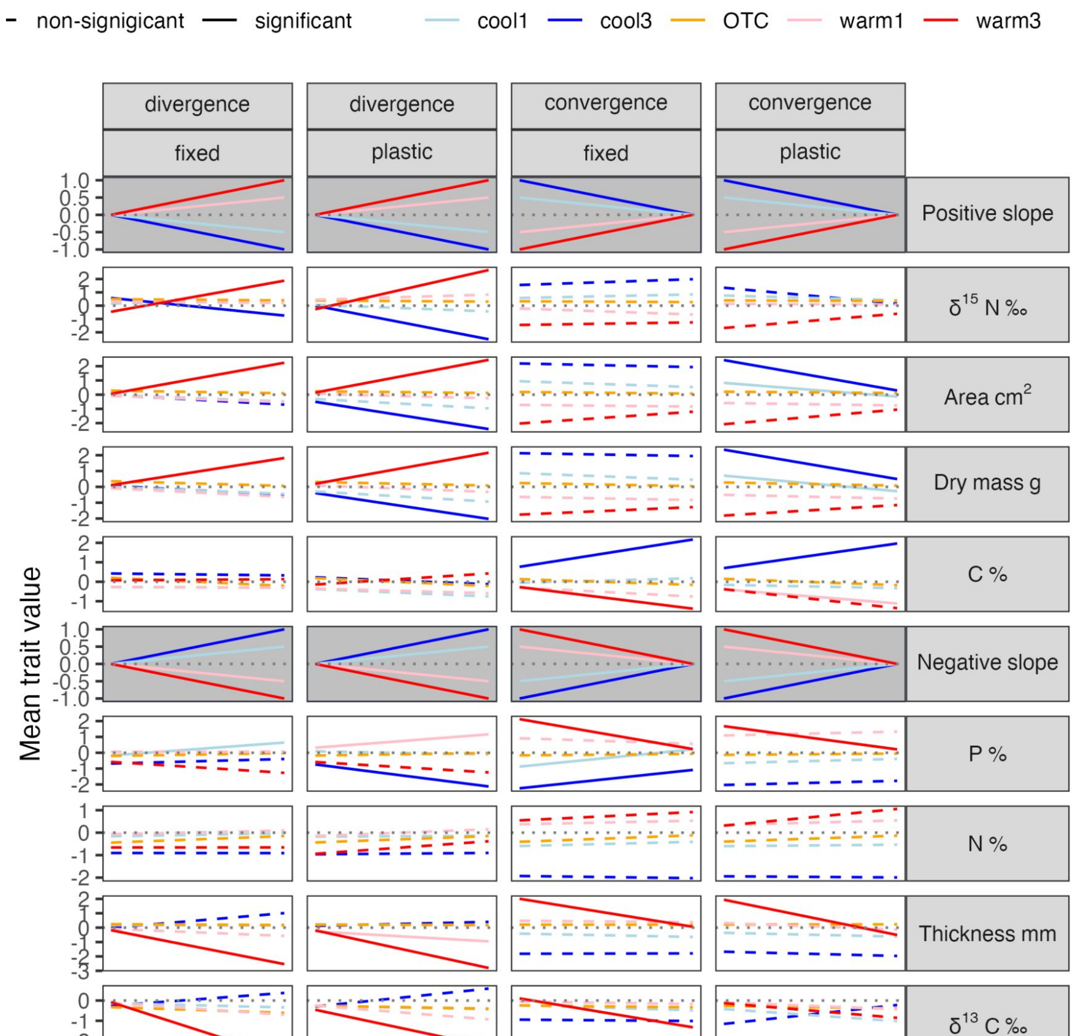


Predicting traits

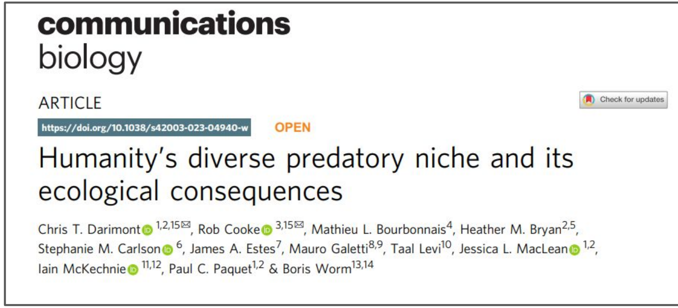
Expected patterns from gradient



Predicting traits



Traits and humans


communications biology

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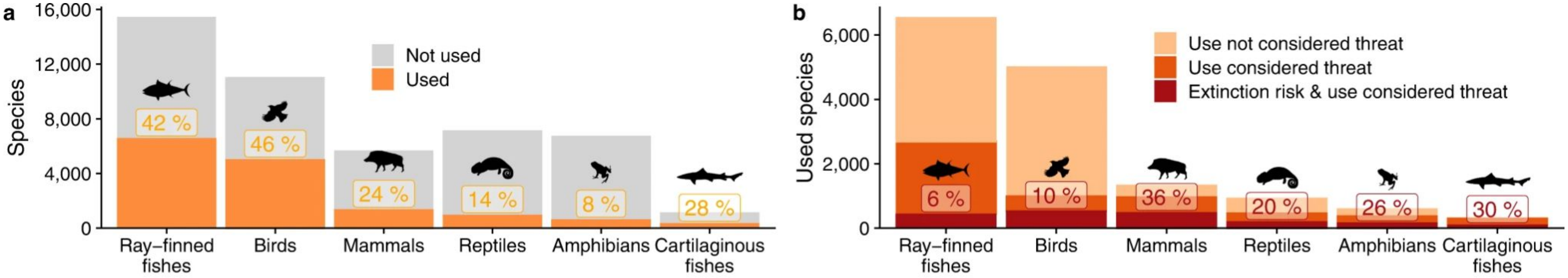
https://doi.org/10.1038/s42003-023-04940-w OPEN

Humanity's diverse predatory niche and its ecological consequences

Chris T. Darimont[1,2,15✉], Rob Cooke[3,15✉], Mathieu L. Bourbonnais[4], Heather M. Bryan[2,5], Stephanie M. Carlson[6], James A. Estes[7], Mauro Galetti[8,9], Taal Levi[10], Jessica L. MacLean[1,2], Iain McKechnie[11,12], Paul C. Paquet[1,2] & Boris Worm[13,14]



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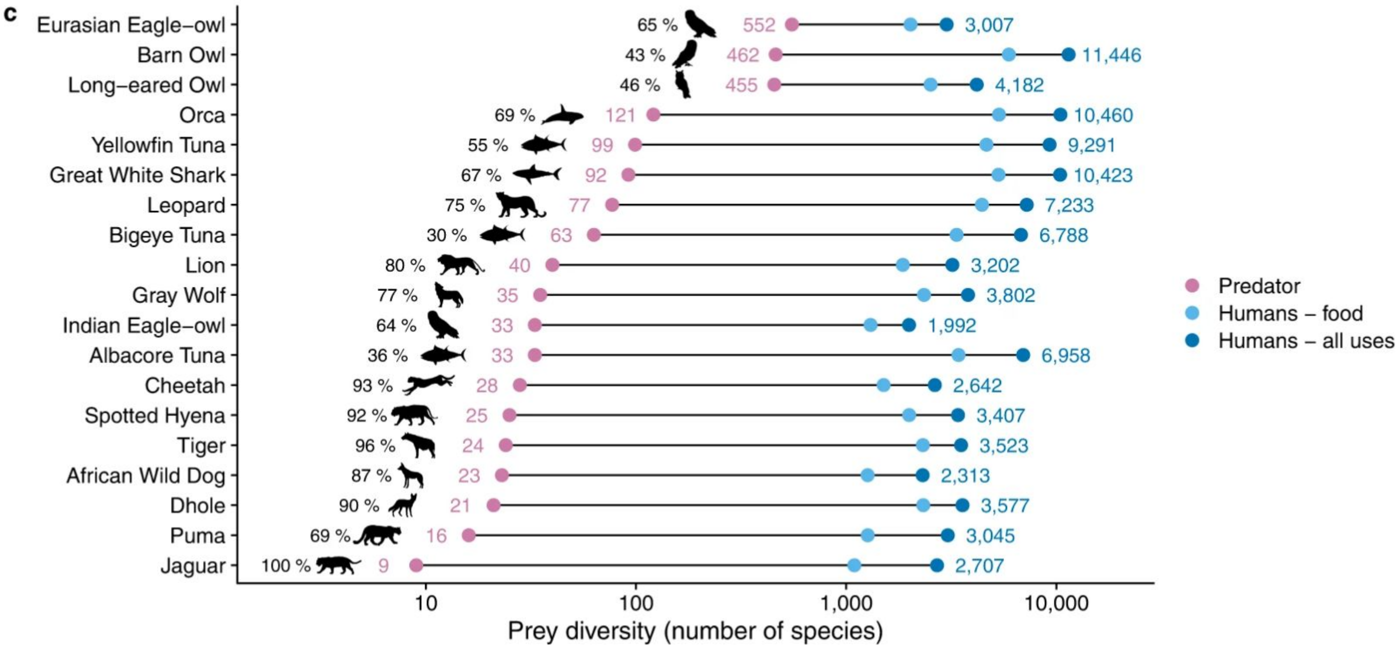

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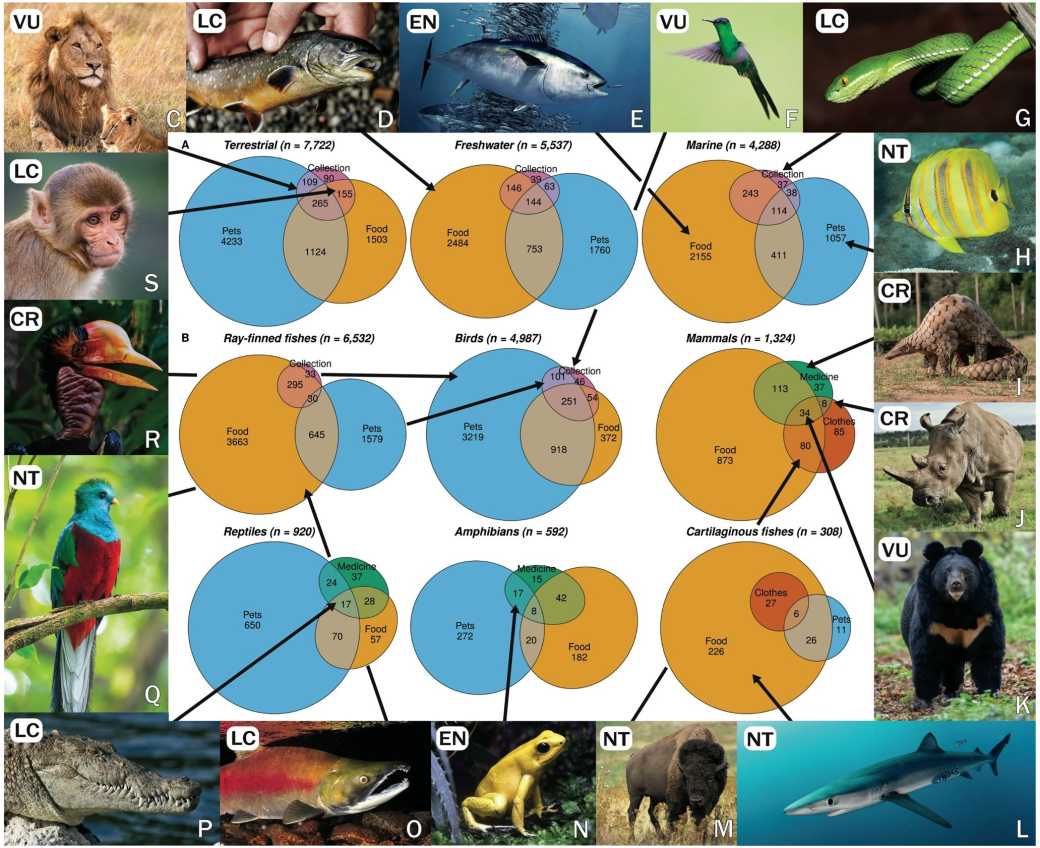

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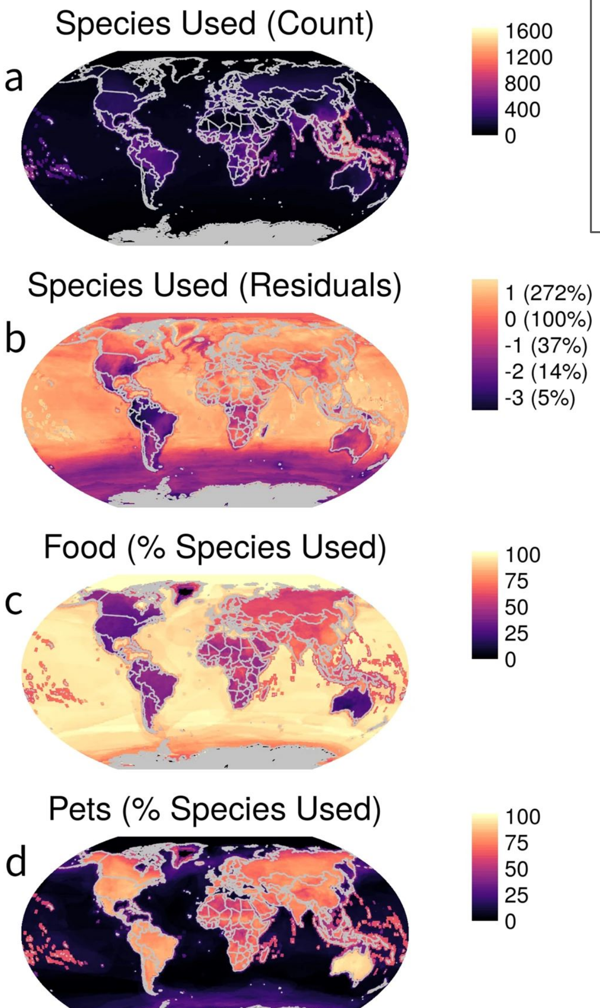

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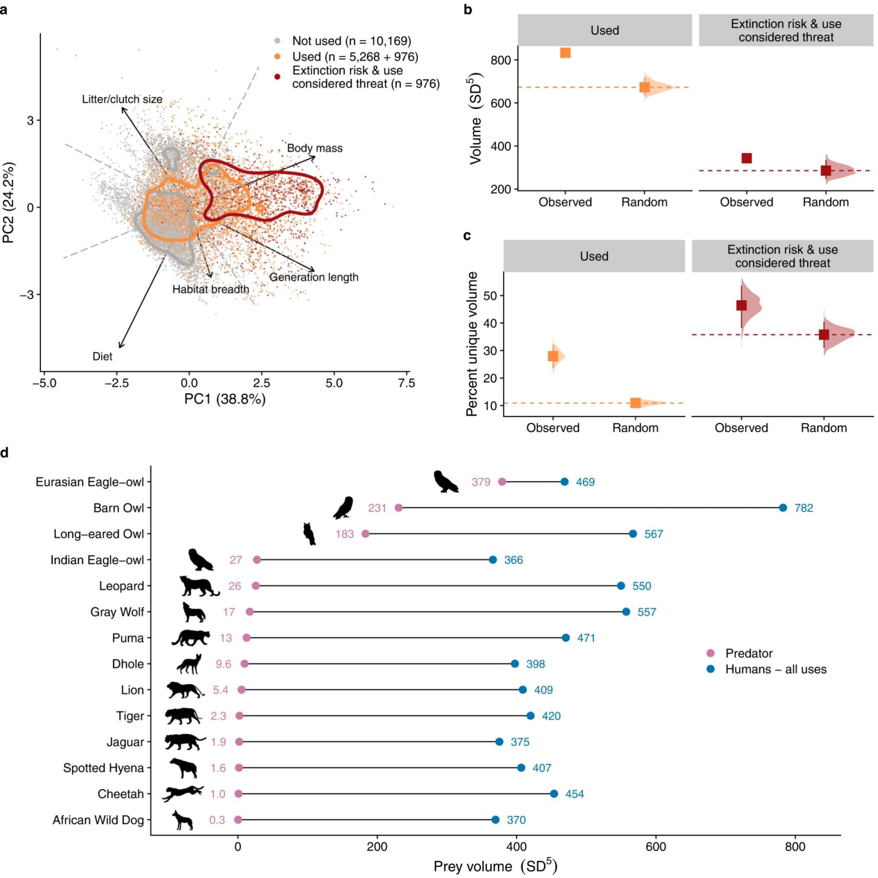

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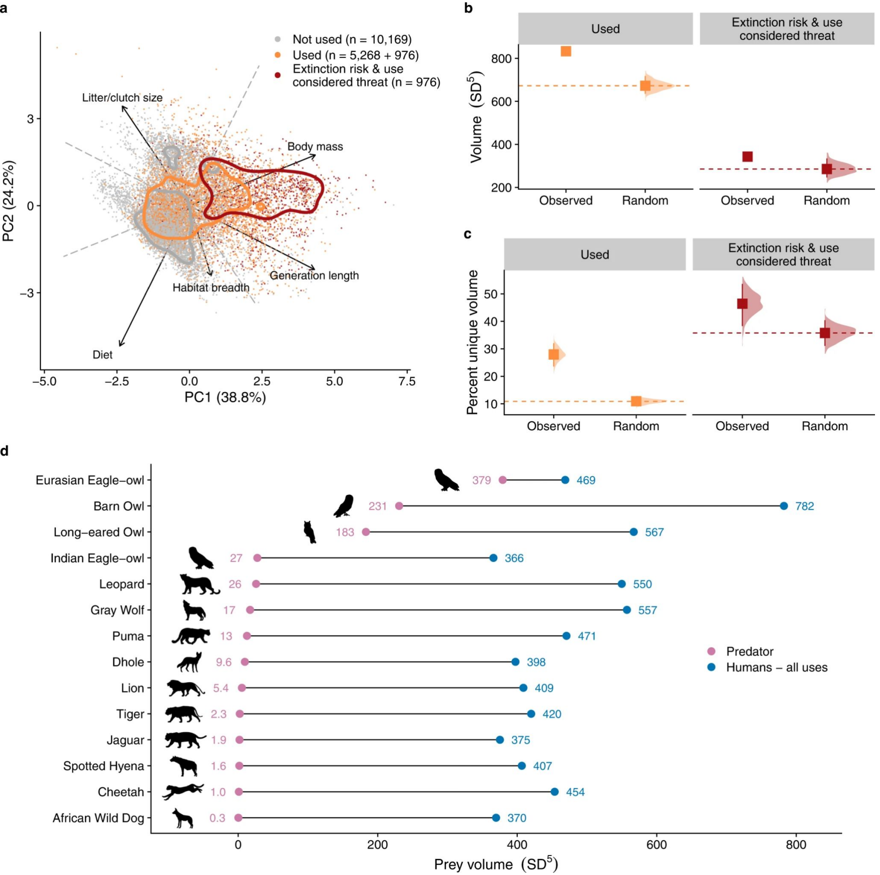

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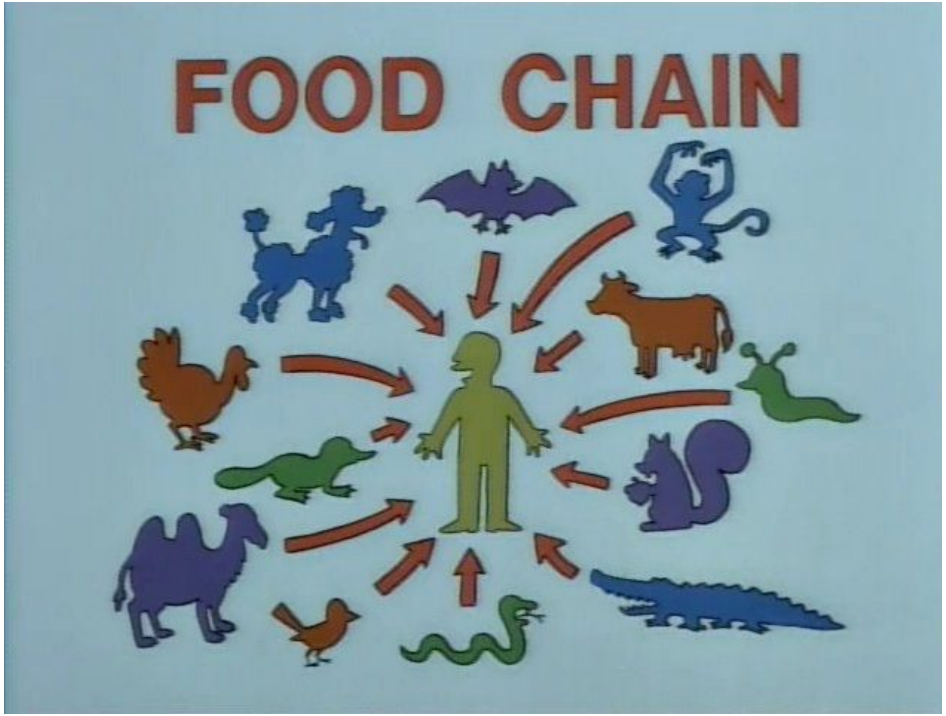


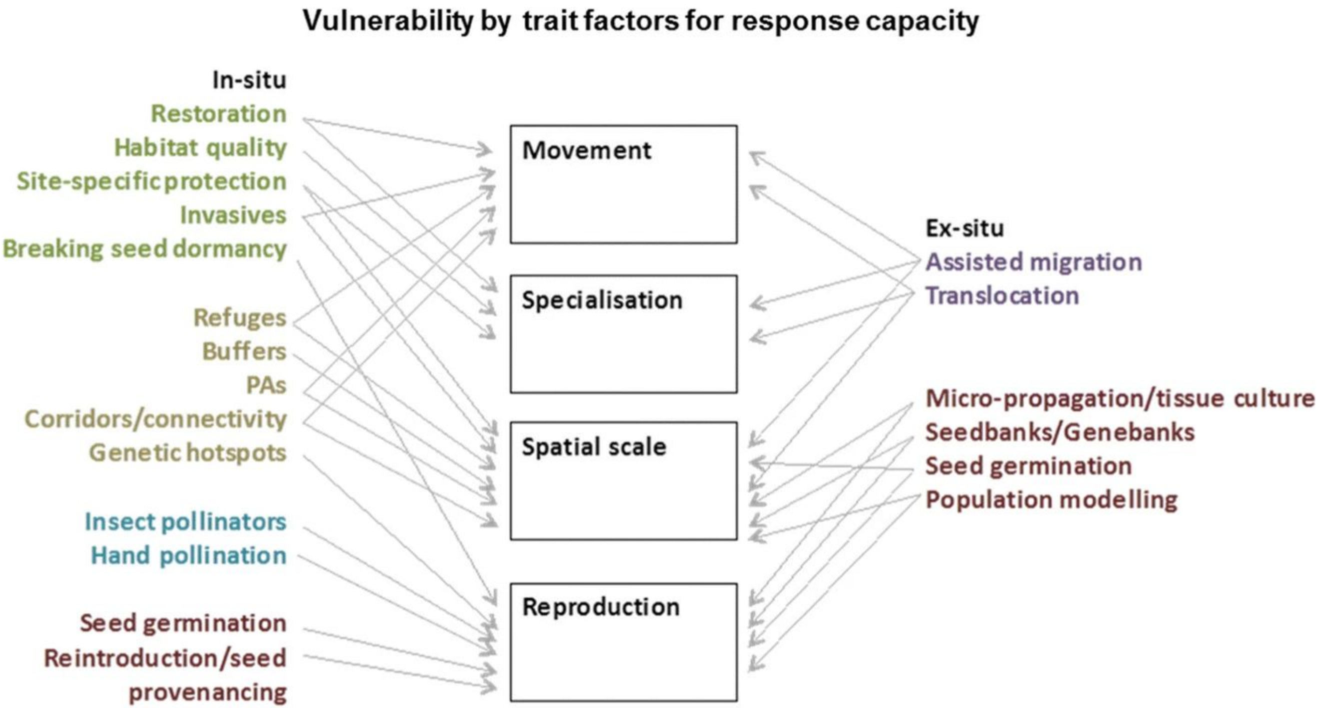
Traits and conservation

Climatic Change (2018) 151:317–332
https://doi.org/10.1007/s10584-018-2294-z

CrossMark

Using species traits to guide conservation actions under climate change

Nathalie Butt[1] · Rachael Gallagher[2]



In-situ and ex-situ **management actions** grouped as: habitat; landscape; interspecific interaction; species-specific; translocation. The arrows indicate which factors the actions can act upon to maximise response capacity

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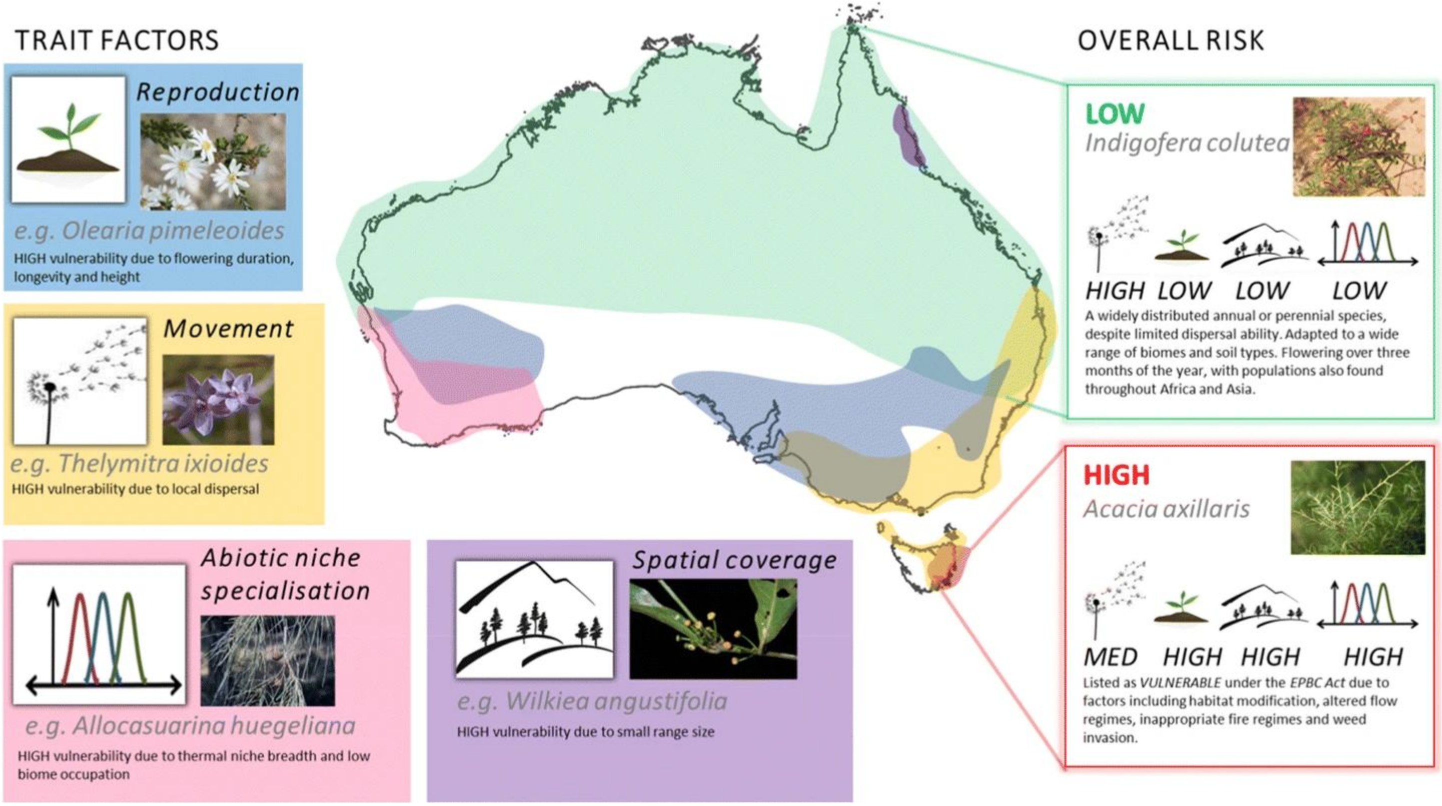

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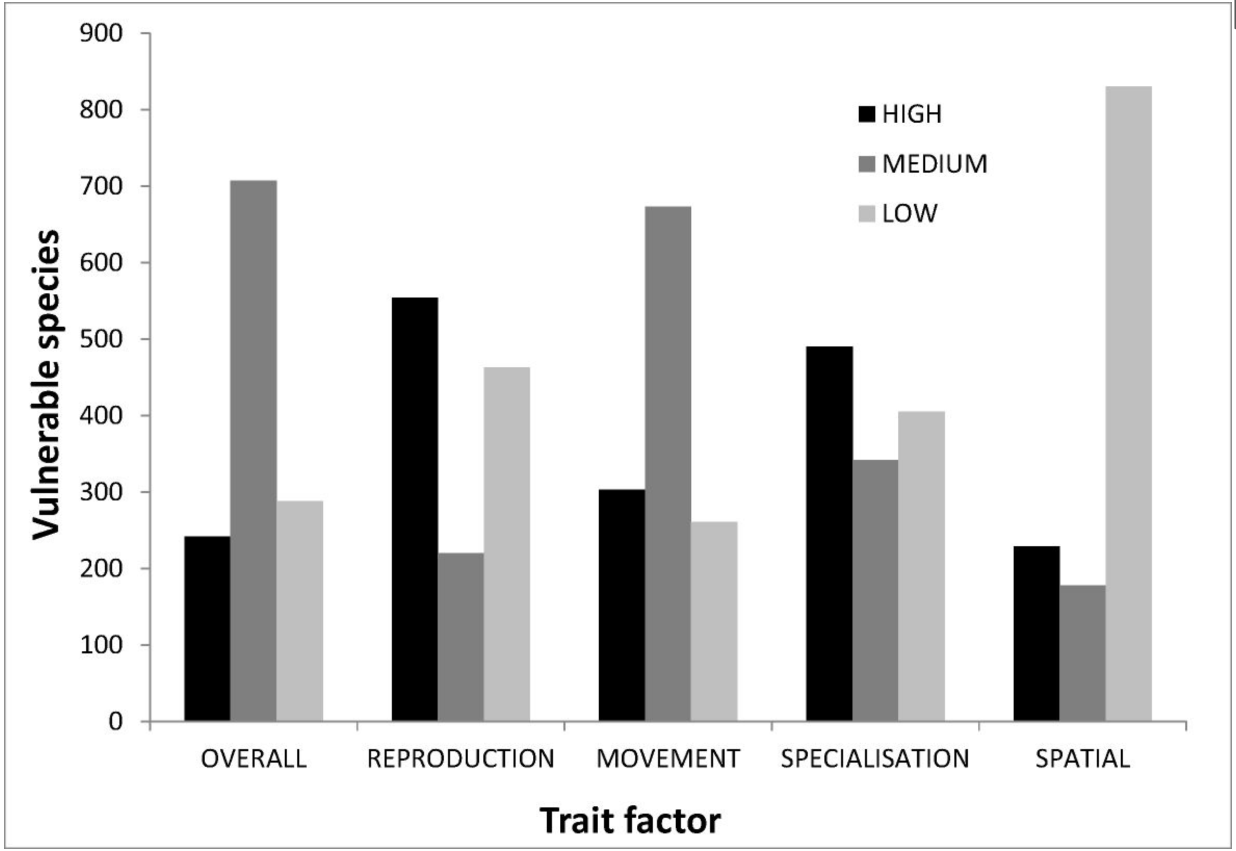

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Traits and conservation

One Earth
Gold Standard for the Global Goals
1 ton of CO_2 offset
CellPress

Perspective

A guide to using species trait data in conservation

Rachael V. Gallagher,[1,*] Nathalie Butt,[2,3] Alexandra J.R. Carthey,[1] Ayesha Tulloch,[2,4] Lucie Bland,[5,6] Simon Clulow,[1,7] Thomas Newsome,[4] Rachael Y. Dudaniec,[1] and Vanessa M. Adams[1,8]

Integrating traits along the conservation continuum

GOALS	**Assessing vulnerability/predicting extinction risk**	**Designing management strategies**	**Prioritising actions**	**Implementing conservation actions across the landscape**	**Monitoring and evaluation of actions**
UTILITY OF TRAITS	Assessing species, populations or communities for sensitivity and adaptive capacity to threats and extinction	Creating approaches to effectively mitigate and manage threats to persistence and survival	Ranking species, populations or communities based on their monitoring and management needs	Spatially prioritising the implementation of complementary conservation actions	Assessing the success or failure of conservation actions designed to prevent extinction
	Traits can… capture responses to threats and disturbances (e.g. for fire – resprouting capacity, climate change – dispersal ability, coral bleaching - resilience)	**Traits can…** inform which actions promote population persistence (e.g. assisted migration may be favoured over *in situ* protection for species with low dispersal capacity)	**Traits can…** act as surrogates for the likely response of species to threats to prioritize which threat management actions will deliver the greatest return on investment	**Traits can…** be used in threat mapping exercises to complement spatial data on exposure to pressure such as climate change and drought risk	**Traits can…** be used to group entities based on appropriate monitoring intervals or their likely responses to management actions
	Examples In birds [22], mammals [23], butterflies [24], moths [25], across multiple taxa [9]	**Examples** In amphibians [26]; freshwater fish [27]; plants and forests [28-29]	**Examples** For reef management [30]; feral animal impacts [31]; post-fire recovery [12]	**Examples** For adressing land-use change [32-33]; designing conservation zones [34-35]	**Examples** In grasslands [36-37]; marine organisms [38-39]; tropical forests [40]

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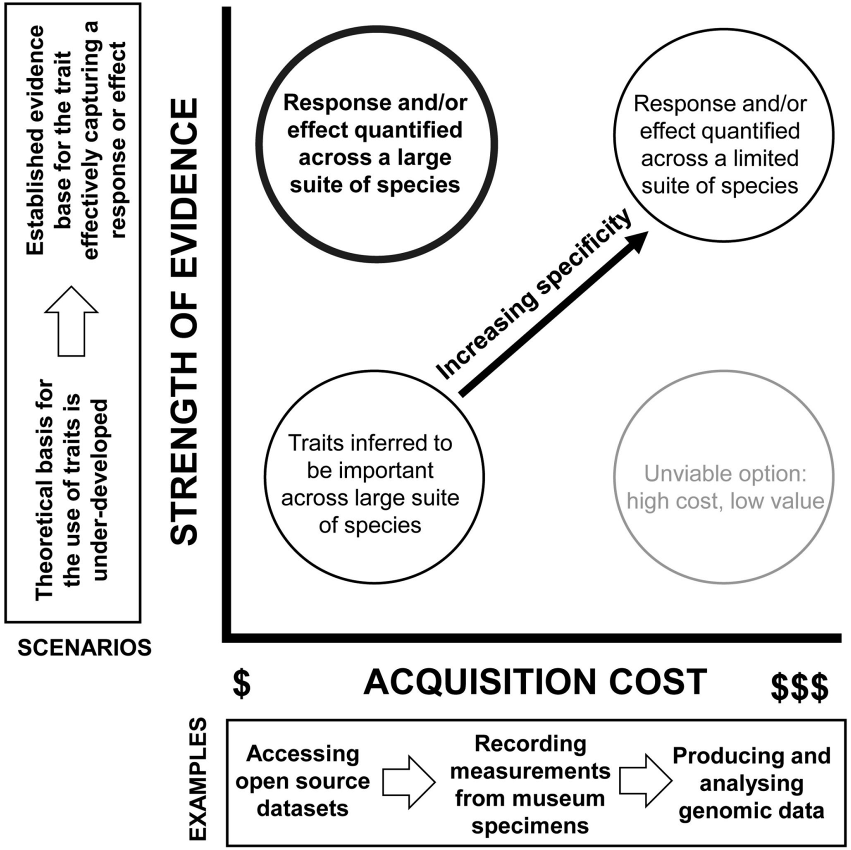

Next class: Patchy environments